## Problem 3

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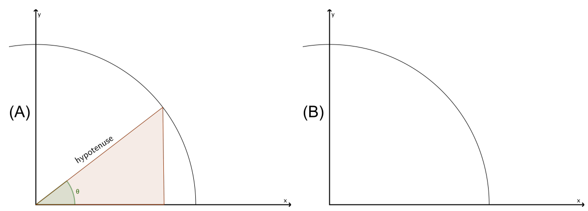
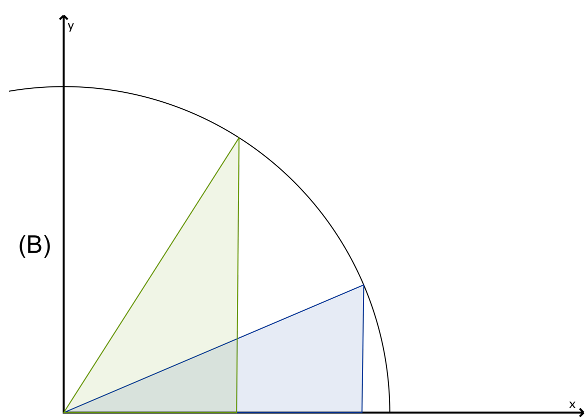
**Problem 1** *A part of a circle centered at the origin with radius $r = 7$ cm is given in the figure (A) below. A right triangle is formed in the first quadrant (see figure (A)). One of its sides lies on the $x$-axis. Its hypotenuse runs from the origin to a point on the circle. The hypotenuse makes an angle $\theta$ with the $x$-axis.*




*Make sure to label the picture.*

(a) *Draw 2 more examples of such a triangle in the figure (B).*

**Explanation.**




(b) *Express the area of such a triangle as a function of $\theta$ and state its domain.*

**Explanation.** The area of a triangle is $(1/2) \cdot$ base $\cdot$ height . The base of the triangle is $7\cos(\theta)$ and the height is $7\sin(\theta)$.

Therefore:

$$\begin{aligned} A(\theta) &= \frac{1}{2} \cdot 7\cos(\theta) \cdot 7\sin(\theta) \\ &= \frac{49}{2}\cos(\theta)\sin(\theta) \end{aligned}$$

and

$$\text{Domain of } A = [0, \pi/2]$$

(c) *Find the value of $\theta$ which maximizes the area in part (b). Show your work and justify your answer. [2]*

**Explanation.** Finding critical points:

$$\begin{aligned} A'(\theta) &= \frac{-49}{2}\sin(\theta)\sin(\theta) + \frac{49}{2}\cos(\theta)\cos(\theta) \\ &\implies A'(\theta) = 0 \\ &\implies \cos^2(\theta) = \sin^2(\theta) \\ &\implies \cos(\theta) = \sin(\theta) \\ &\implies \theta = \frac{\pi}{4} \end{aligned}$$

Locating global maximum:

$$A(0) = 0, A(\pi/4) = \frac{49}{4}, A(\pi/2) = 0 \implies \text{global maximum at } \theta = \pi/4$$